Roll No. | | | | | | | | | | |

Total No. of Pages : 02

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B.Pharmacy (Sem.-6)

MEDICINAL CHEMISTRY-III-THEORY

Subject Code : BP-601T

M.Code : 77986

Date of Examination : 06-05-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. Answer briefly :

a) Give chemical structure and uses of Penicillin G.

b) Write the structure and uses of Chloramphenicol.

c) What is lead molecule? How are they useful in drug discovery?

d) Write synthesis of Miconazole.

e) Define Taft's steric constant of substituent. Comment on its applicability.

f) Write chemical structure and uses of Amantadine.

g) Write chemical structure and uses of Methanamine.

h) What are beta lactamase inhibitors?

i) Which alkaloid has anti-malarial activity?

j) Write the structure and uses of Zidovudine.

SECTION-B

2. Give SAR of quinolones used as urinary tract anti-infective agents. Write synthesis of ciprofloxacin.

3. Write the SAR and chemistry of sulpha drugs. Write the synthesis of Trimethoprim.

4. Write a note on prodrug concept and combinational chemistry in drug discovery.

SECTION-C

5. What is fundamental principle of QSAR? Discuss any one conventional 2D QSAR model in detail.

6. Write a short note on HIV protease inhibitors by giving suitable examples.

7. What are antifilarial drugs? Discuss in detail Ivermectin.

8. What are beta lactam antibiotics? Write the degradation products of penicillin.

9. What are anti-protozoal agents? Give the structures of Iodoquinol and Metronidazole.

10. Write chemical structure of Tetracycline, Chlortertracycline. Mention their three therapeutic uses.

11. Classify the cephalosporins based on generation. Write the structure and uses of Cephalexin.

12. What are macrolides? Write their mode of action. Give structure and medicinal uses of erythromycin.

13. Define and classify anthelmintics. Write the synthesis of Mebendazole.

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NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.